

Chabala Vineeth Babu

6300633356 | vineeth.chabala | linkedin.com/in/vineethchabala/ | github.com/VineethChabala

EDUCATION

IIITDM Kancheepuram

Bachelor of Technology, Computer Science and Engineering

8.3/10.0

2020 – 2024

PROFESSIONAL SUMMARY

Software Engineer specializing in data-intensive IoT platforms and production-grade Machine Learning. Expertise in building scalable end-to-end systems using **Python**, **Flask**, **Kafka**, and **Kubernetes**. Proven track record in developing deep learning models (TensorFlow/Keras) and optimizing algorithms by up to **76%**.

EXPERIENCE

Software Engineer

Agneyas Labs

Jan 2025 - present

Bangalore, India

- **Architected** an end-to-end IoT platform for real-time EV battery diagnostics, transitioning the system from concept to a production-ready environment. **Developed** a high-throughput data pipeline using Kafka to stream hardware telemetry from Python controller services to a Flask backend. **Implemented** real-time monitoring by delivering live battery metrics to an Angular dashboard via **Websockets**.
- **Engineered** a Python-based core controller service to interface with physical hardware, interpreting **CAN bus** signals and automating diagnostic sequences via Relays and DMMs.
- **Built** a comprehensive simulation ecosystem in Python for EV Batteries and hardware components, accelerating the development lifecycle by enabling hardware-independent validation.
- **Orchestrated** the deployment of microservices using **Docker** and **Kubernetes**, ensuring a scalable and resilient production infrastructure.
- **Engineered a consolidated EV Teststation dashboard**, enabling simultaneous multi-station monitoring and centralized data visualization for streamlined hardware testing operations .
- **Leading the migration** of the company's enterprise source code from **Angular 7 to Angular 20**, modernizing frontend architecture and leveraging latest framework features for improved performance.
- **Orchestrating** the transition of the backend infrastructure from **Python 3.7 to 3.10**, utilizing new language features like structural pattern matching and improved type hinting to increase code maintainability.

Full Stack Development Intern

Agneyas Labs

June 2024 - Jan 2025

Bangalore, India

- **Designed** and implemented a RESTful backend for a real-time work order scheduling system using **Flask** and **Python**.
- **Optimized** data persistence by utilizing **InfluxDB** for high-volume time-series data, ensuring high-performance retrieval for real-time machine availability visualization.
- **Enhanced** frontend usability by developing a reusable chip-based selection component in **Angular 7**, which was subsequently adopted as the organizational UI standard.

PROJECTS

Moodie AI | *Python, Streamlit, TMDB API*

Jan 2026

- **Developed** a context-aware movie recommendation engine that maps complex human emotional states and "energy levels" to precise **REST API** queries.
- **Implemented** advanced session state management to handle persistent user data, enabling seamless "skip/seen" logic without reloading the application.
- **Engineered** a multi-factor filtering algorithm that adapts recommendations based on social context (e.g., "Date Night" vs. "Family") and historical era preferences.
- **Designed** an interactive UI using **Streamlit**, integrating real-time data fetching for movie posters, metadata, and popularity metrics.

Solving Equal Piles Problem | *Python*

May 2024

- **Developed** a hybrid optimization model combining a **Genetic Algorithm** with **Perturbed Local Search (PLS)** to solve complex data distribution challenges.

- **Achieved a 76% performance increase** over standard PLS methods in balancing uniform data distributions.
- **Benchmarked** model robustness against HGGGA and PLS across **14 unique test cases**, consistently minimizing pile inequality across diverse data scales.

Aquatic Animal Classification | *Tensor Flow, Keras, Python*

November 2023

- **Collaborated** on a computer vision suite evaluating **custom CNN, MobileNet V2, and ResNet50V2** architectures for multi-class classification.
- **Architected the Flask deployment server**, enabling real-time model inference for the final system.
- **Optimized** training parameters for a custom CNN, contributing to a final model accuracy exceeding **82%**.

Operator I | *Flutter, Dart, Flask, Python*

May 2023

- **Engineered** a cross-platform mobile application to interface with a specialized **REST API** for remote image processing.
- **Implemented** secure client-side logic for local file permission management and asynchronous data rendering.
- **Owned** the end-to-end mobile development lifecycle, from API integration to UI/UX delivery.

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, JavaScript, HTML/CSS, TypeScript, Dart, Go

Frameworks & Libraries: Streamlit, Angular, Flutter, Flask, NumPy, Pandas, TensorFlow, PyTorch, scikit-learn, Keras, OpenCV, Matplotlib,

Specialisations: Deep Learning (CNNs), Genetic Algorithms, Local Search Optimization, RESTful API Integration, State Management

Databases: MySQL, InfluxDB, SQLite3, MSSQL

DevOps & Infrastructure: Docker, Kubernetes, Nginx, uWSGI, Kafka

Software: Logisim, AutoCAD, Arduino IDE, VS Code, Git

LEADERSHIP AND VOLUNTEER EXPERIENCE

- Led a team of 30 students to execute promotional campaigns for all institute-level technical events, resulting in a 20% increase in fest attendance.
- Led the publicity team responsible for the Institute's technical clubs, driving student engagement by managing all social media content and serving as the primary video editor for promotional materials.
- Drove content strategy and coordinated publicity campaigns for VIDHAI, a non-profit student initiative, to help underprivileged students study for competitive exams.